AURA MULTI-SPECIALTY HOSPITAL

COMPLETE HOSPITAL MANAGEMENT SYSTEM (HMS) TRAINING & OPERATIONS MANUAL

From Beginner to Professional: Architecture, Accounts, Modules, Workflows, and Every Button Explained

Target Audience:	Hospital Owners, Administrators, Medical Officers, Software Demonstrators & Trainees
Scope of Document:	All 21 User Roles, 27 Core Modules, Every Operational Button, Double-Entry GL, Live Demo Script & Glossary
Core Technologies:	React + TypeScript Frontend, Django REST Framework Backend, Supabase PostgreSQL, Render Cloud
Compliance:	NABH / NABL Clinical Audit Trails, ICD-10 Coding, FEFO Drug Dispensing, Role-Based Access Control (RBAC)
Currency & Locale:	Indian Rupees (■), Multi-Branch Support, 5-Language Interface Synchronization

Prepared for Hospital Demonstration, User Training & Technical Presentation

Version 3.4 • All Systems Connected • Zero Artificial Dummy Transactions

SECTION 1: Introduction & High-Level Project Overview

What is a Hospital Management System (HMS)?
In a real hospital, hundreds of critical events occur simultaneously: outpatient walk-ins, emergency trauma admissions, doctor clinical consultations, medication charting, blood test analysis, medical imaging, pharmaceutical inventory management, cashier reconciliations, and vendor procurement. Without a unified digital system, hospitals suffer from lost medical paper records, billing discrepancies, dangerous drug-allergy interactions, and chaotic patient queues. AURA HMS serves as the centralized digital backbone connecting all hospital branches and departments into a single secure platform.

The Three Foundational Pillars of the HMS:

1. **Patient Care & Front Desk:** Master Patient Index (MRN generation), OPD appointment scheduling, mandatory advance payment collection, queue token generation, and Digital Waiting Lounge TV broadcasting.
2. **Clinical & Diagnostic Services:** Electronic Medical Records (EMR), Doctor Consultation Cabins, Nursing Station Medication Administration Records (MAR), Diagnostic Pathology (LIS), Medical Imaging (RIS), Operation Theatre (OT), and ICU telemetry.
3. **Finance & Logistics Operations:** First-Expiring, First-Out (FEFO) pharmacy dispensing, purchase order procurement, hospital warehouse inventory, cashier shift reconciliation, double-entry General Ledger accounting, HR employee rosters, and monthly payroll.

SECTION 2: Detailed Account Roles & Permission Matrix

Every hospital staff member has a dedicated account governed by strict Role-Based Access Control (RBAC). Below is the complete directory of all 21 roles in the system:

Role Name	Real-Hospital Designation	Key Responsibilities & Allowed Actions	Strict Prohibitions
SUPER_ADMIN	Chief System Administrator	Full global authority across all branches. System configuration, provisioning staff accounts, master RBAC overrides, security audit inspection.	Cannot delete finalized clinical medical records or past journal vouchers.
HOSPITAL_ADMIN	Hospital Administrator / COO	Central pricing & tariff management across all branches, service catalog rates, departmental operations oversight, facility management.	Strictly blocked from modifying doctor clinical consultation notes or dispensing pharmacy stock.
BRANCH_MANAGER	Branch Operations Manager	Supervisory read-only command center across 14 operational domains (OPD, IPD, OT, LIS, RIS, Billing, Pharmacy, Staff Roster).	Strict read-only enforcement (HTTP 403 on all write attempts). Cannot modify invoices, clinical notes, or tariffs.
MEDICAL_DIRECTOR	Chief Medical Officer (CMO)	Clinical governance, medical protocol review, doctor quality metrics, mortality review, clinical audit trails.	Cannot alter financial general ledgers or bank reconciliation records.
DEPARTMENT_HEAD	Clinical HOD (e.g. Cardiology)	Departmental doctor roster scheduling, shift swap reviews, staff requirement requests, clinical performance reviews.	Cannot provision login credentials or access other clinical departments.
DOCTOR	Consultant / Attending Physician	Conducts OPD consultations, records vitals and ICD-10 diagnoses, prescribes medications (e-Rx), orders lab/radiology tests, signs discharge summaries.	Strictly blocked from booking appointments, collecting billing cash, or editing locked historical visits.
NURSE	Staff Nursing Officer	Ward inpatient care, hourly vital logs, Medication Administration Record (MAR), nursing handover notes (SBAR), nursing discharge clearance.	Cannot prescribe new medications or alter financial invoices.
RECEPTIONIST	Front Desk Executive	Patient registration, OPD appointment scheduling, mandatory consultation payment collection, token issuance, queue coordination.	Strictly blocked from entering clinical notes or altering pharmacy inventory.

BILLING_EXECUTIVE	Inpatient & Outpatient Billing Staff	Generates itemized patient bills, discharge settlements, advance deposits, split payments (Cash, Card, UPI), prints official tax invoices.	Cannot approve unverified billing discounts without administrative authorization.
CASHIER	Cash Counter Executive	Cash drawer management, daily opening float entry, cash collection, shift closure cash reconciliation, physical count audit.	Cannot modify clinical diagnosis or alter item prices on the fly.
PHARMACIST	Registered Pharmacist	Receives doctor-signed e-prescriptions, validates drug interactions, dispenses medications using FEFO, manages batch expiries.	Cannot dispense expired batches or dispense without verified doctor prescription.
INVENTORY_MANAGER	Materials & Stores Manager	Central hospital warehouse, stock movements, batch tracking, stock write-offs, minimum reorder thresholds.	Cannot authorize vendor payments directly (requires Accounts separation).
PROCUREMENT_OFFICER	Hospital Purchasing Officer	Prepares Purchase Requisitions, solicits vendor quotations, prepares comparative price charts, generates Purchase Orders (POs), logs Goods Received Notes (GRN).	Strictly prohibited from self-approving purchase orders without separation of duty.
LAB_TECHNICIAN	Medical Laboratory Technician	Specimen accessioning, sample barcode labeling, analyzer result entry, critical panic value alerts, report submission for pathologist sign-off.	Cannot release diagnostic reports without medical officer digital authorization.
RADIOLOGY_TECHNICIAN	Radiographer / Imaging Tech	Executes imaging studies (X-Ray, CT, MRI, Ultrasound), attaches PACS/DICOM study links, prepares draft technical impressions.	Cannot authorize clinical radiological interpretations without Radiologist verification.
FINANCE_MANAGER	Chief Financial Officer / Finance Head	Authorizes high-value vendor payments, posts double-entry journal vouchers, reviews Profit & Loss (P&L;), manages corporate bank accounts.	Cannot alter verified patient clinical histories.
ACCOUNTANT	Senior Hospital Accountant	Prepares journal entries, reconciles bank statements, tracks Accounts Receivable (AR) and Accounts Payable (AP), audits cashier drawers.	Cannot approve self-prepared vouchers without supervisor authorization.
INVESTMENT_MANAGER	Treasury & Investment Officer	Manages hospital fixed deposits (FD), government bonds, treasury reserves, valuation logs, interest accruals, investment liquidation.	Cannot divert hospital funds outside approved corporate treasury policies.
INSURANCE_COORDINATOR	TPA Desk Executive	Patient insurance verification, cashless pre-authorization, TPA query resolution, claim document submission, settlement reconciliation.	Cannot mark bills settled without verifiable TPA approval sanction letters.
HR_MANAGER	Human Resources Head	Onboards new staff, provisions approved operational login accounts, manages leave policies, attendance regularizations, payroll runs.	Cannot provision high-governance administrative roles (SUPER_ADMIN, HOSPITAL_ADMIN, BRANCH_MANAGER).
PATIENT	Registered Hospital Patient	Accesses self-service portal: views past appointments, electronic prescriptions, verified diagnostic reports, invoices, and payment receipts.	Strictly restricted to own records only; zero visibility into other patients or hospital internal modules.

SECTION 3: The 27 Core Modules – Purpose, Fields & Functionality

1. Master Patient Index (MPI)

Maintains universal demographic records, contact details, emergency contacts, blood group, allergies, and generates the permanent Medical Record Number (MRN-YYYY-XXXXX).

2. OPD Appointments & Token Queue

Doctor scheduling, time-slot allocation, advance consultation fee collection, token generation, and multi-counter queue management.

3. Digital Waiting Lounge TV Display

Public audiovisual queue announcer broadcasted on waiting room screens showing current tokens being called into doctor chambers.

4. Doctor Consultation Cabin (EMR)

Clinician workspace: vital signs logging, Chief Complaints, past medical history, ICD-10 diagnosis, e-Prescriptions with allergy alerts, and electronic test orders.

5. Nursing Station & MAR

Ward-level inpatient monitoring: vital sign charts, Medication Administration Record (due, administered, missed, refused doses), and structured SBAR shift handovers.

6. Inpatient (IPD) Bed Census

Interactive visual bed board managing General, Semi-Private, Deluxe, and ICU beds with status indicators (Available, Occupied, Cleaning, Maintenance).

7. Emergency & Trauma Triage

Rapid arrival triage sorting emergency patients into Red (Resuscitation), Yellow (Urgent), and Green (Non-urgent) priority streams.

8. Operating Theatre (OT) Suites

Surgery scheduling, surgical team rostering (Lead Surgeon, Anesthetist, Scrub Nurse), pre-op checklists, and anesthesia recovery monitoring.

9. Intensive Care Unit (ICU)

Critical care telemetry tracking invasive ventilator settings, continuous SpO2/BP monitoring, Glasgow Coma Scale (GCS), and organ failure scores.

10. Laboratory Information System (LIS)

Requisition intake, specimen accessioning, barcode generation, analyzer result entry, reference range comparison, critical panic flags, and pathologist sign-off.

11. Radiology Information System (RIS)

Imaging examination scheduling, DICOM study metadata linkage, modality worklist, and structured radiologist impression reports.

12. Pharmacy Formulary & Dispensing

Direct intake of doctor e-prescriptions, FEFO inventory allocation, drug interaction checks, batch expiry blocking, and patient dispensing.

13. Multi-Branch Procurement

Complete supply chain lifecycle: Department Purchase Requisitions -> Vendor Quotations -> Comparative Statements -> Purchase Orders (POs) -> Goods Received Notes (GRNs).

14. Central Inventory & Warehouse

Warehouse management tracking on-hand batch balances, storage locations, minimum safety stock alerts, and audited stock write-offs.

15. Billing, Invoicing & Cashier Shifts

Comprehensive billing combining bed days, doctor fees, lab, radiology, and pharmacy items. Manages cashier drawer opening/closing floats and collections.

16. Insurance & Corporate TPA Claims

Cashless hospitalization tracking, pre-authorization requests, insurer query management, claim document dockets, and settlement vouchers.

17. Central Pricing & Tariff Management

Hospital Admin authority to centrally configure consultation fees, bed tariffs, and clinical service charges with synchronized 1-click push across all branches.

18. Double-Entry General Ledger (GL)

Full financial accounting engine where every transaction generates balanced debit and credit entries. Immutable audit trails with reversal capability.

19. Profit & Loss (P&L;) Statement

Real-time revenue, direct operational costs, staff expenses, and net hospital surplus calculation derived strictly from posted ledger transactions.

20. Cash Flow & Bank Accounts

Multi-branch bank account balances, cash-in-hand tracking, electronic fund transfers, cheque clearing, and bank reconciliation statements.

21. Fixed Assets & Depreciation

Hospital medical machinery (MRI, X-Ray, Ambulances) tracking, preventive maintenance tasks, breakdown work orders, and straight-line annual depreciation.

22. Hospital Investment Registers

Corporate treasury management for Fixed Deposits, government bonds, interest yield tracking, maturity alerts, and liquidity management.

23. HR Operations & Duty Rosters

Staff profiles, department rosters, shift swap requests, leave applications, biometric punch regularizations, and credential verification.

24. Payroll & Salary Disbursal

Monthly payroll generation, basic pay, house rent allowance (HRA), provident fund (PF) statutory deductions, 1.5x overtime multiplier, and payslip generation.

25. Branch Manager Supervisory Cockpit

High-level operational overview across 14 domains with strict read-only permissions preventing operational record tampering.

26. Patient Self-Service Portal

Secured web portal allowing patients to access their medical summaries, discharge dockets, lab reports, and payment history from home.

27. Security, Audits & Activity Logs

Hospital-wide tamper-proof activity logging recording every user action, IP address, timestamp, affected entity, and permission denials.

SECTION 4: Comprehensive Button & Action Directory

Below is the exact operational directory explaining what happens behind every primary button in the application:

Button Name & Location	Who Can Click?	Action Performed & Data Entered	Post-Click Behavior & System Impact
Register Patient Front Desk / Patients Page	Receptionist, Admin	Enters name, gender, DOB, phone, blood group, emergency contact, allergies.	Saves to PostgreSQL `PatientProfile`. Generates unique `MRN-YYYY-XXXXX`. Immediately selectable in appointment and admission forms.
Book Appointment Appointments Page	Receptionist, Patient	Selects branch, department, consulting doctor, date, and time slot.	Locks appointment slot. Prompts mandatory payment collection dialog. Doctors are forbidden from booking.
Collect Payment & Issue Token Appointment Modal	Receptionist, Cashier	Selects payment mode (Cash, Card, UPI). Enters collected amount.	Creates payment transaction, posts debit to Cash/Bank account, generates sequential queue token (e.g. #A-014), and prints receipt.
Call Patient Doctor Consultation Workspace	Doctor	Clicks token in active waiting queue.	Plays digital audio chime on Waiting Lounge TV, flashes cabin number, marks appointment status as IN_CONSULTATION.
Finalize Consultation Doctor Workspace	Doctor	Reviews diagnosis (ICD-10), vital signs, prescriptions, and lab orders.	Cryptographically locks consultation note against post-visit edits. Transmits e-Rx to pharmacy and test orders to LIS/RIS.
Admit Patient Admissions & Ward Bed Board	Receptionist, Admission Staff	Selects patient MRN, admitting doctor, ward category, bed number, and advance deposit amount.	Changes bed status from AVAILABLE (Green) to OCCUPIED (Red). Creates `InpatientAdmission` record. Generates advance deposit receipt.
Transfer Bed Inpatient Admission Detail	Admission Staff, Ward Nurse	Selects new destination ward and vacant bed. Enters clinical transfer reason.	Atomic reallocation: frees previous bed (marked CLEANING), occupies new bed, updates prorated running bed tariffs.
Log Medication (MAR) Nurse Workspace	Nurse	Selects scheduled medication dose: Administered, Missed, or Refused. Enters nurse notes.	Logs digital signature, administering timestamp, and updates patient clinical care docket.
Accession Sample Laboratory Orders Queue	Lab Technician	Verifies specimen tube container, collection time, and phlebotomy notes.	Generates unique specimen barcode. Marks order status as SAMPLE_COLLECTED / IN_PROCESSING.
Verify & Release Report Laboratory Report Screen	Pathologist / Medical Officer	Reviews entered test values against biological reference intervals and panic limits.	Signs report digitally. Instantly publishes report to Doctor EMR workspace and Patient Portal.
Dispense Prescription Pharmacy Dispensing Queue	Pharmacist	Scans medicine barcode. System auto-allocates nearest-expiry batch (FEFO).	Validates on-hand stock and expiry dates. Deducts inventory quantity, logs `StockMovement`, and creates pharmacy billing charge.
Initiate Discharge Inpatient Admissions	Attending Doctor	Completes clinical discharge summary, home prescriptions, and follow-up advice.	Starts mandatory 4-Point Clearance workflow. Patient cannot leave until Nursing, Pharmacy, and Billing clear the account.
Settle & Print Final Invoice Billing Desk	Billing Executive, Cashier	Aggregates bed days, doctor fees, lab, and pharmacy charges. Deducts advance deposits. Collects balance or processes refund.	Issues tax invoice with zero balance due. Grants final billing clearance. Bed status transitions to CLEANING.

Apply Central Tariffs Pricing & Tariff Management	Hospital Admin	Edits consultation fee or bed rate. Reviews Impact Analysis preview.	Pushes new tariffs across all hospital branches simultaneously. Past invoices and ongoing admissions preserve contracted rates.
Post Journal Voucher General Ledger Page	Finance Manager, Accountant	Enters balanced debit and credit ledger rows (`Total Debits == Total Credits`) with narration.	Posts immutable journal entry. Updates General Ledger, Trial Balance, and Profit & Loss statement in real time.
Reconcile Cashier Shift Billing Cash Drawer	Cashier, Finance Manager	Enters physical currency denomination count at end of shift.	Compares physical cash against system collections. Flags cash surplus/shortage and locks cashier shift drawer.

SECTION 5: End-to-End Hospital Workflows & Data Connections

Workflow 1: Complete Patient Journey (OPD to IPD Discharge)

1. **Intake:** Patient arrives at reception -> Registered in Master Patient Index -> Permanent MRN generated.
2. **Booking & Payment:** OPD appointment booked with Dr. Ananya -> Mandatory consultation fee (■500) collected in cash -> Queue Token (#A-014) issued.
3. **Waiting Broadcast:** Lounge TV plays audio chime and directs Token #A-014 to Cabin 102.
4. **Consultation:** Doctor records vitals, logs Chief Complaints, selects ICD-10 diagnosis, prescribes antibiotic, orders CBC lab test, and recommends IPD admission.
5. **Admission & Deposit:** Receptionist allocates Bed G-04 in General Ward (■1,500/day). Patient deposits ■5,000 advance. Bed turns OCCUPIED.
6. **Nursing Care:** Nurse monitors vitals hourly and records administered antibiotic injection on the Medication Administration Record (MAR).
7. **Diagnostics:** Lab collects blood, enters CBC values, pathologist digitally verifies report -> Doctor reviews report on screen.
8. **4-Point Discharge:** Doctor completes discharge summary -> Nurse clears bed custody -> Pharmacy accepts unused medicine returns -> Billing calculates total charges (■3,400) and refunds ■1,600 excess advance in cash -> Bed marked CLEANING.

Workflow 2: Pharmacy Inventory & FEFO Dispensing

1. **Electronic Prescription Receipt:** When a doctor finalizes a consultation, the prescription is securely pushed to the pharmacy dispensing queue.
2. **First-Expiring, First-Out (FEFO) Allocation:** The pharmacy engine automatically matches the prescribed drug code with inventory batches and selects the batch closest to expiration.
3. **Dispensing Validation:** The system blocks dispensing if on-hand quantity is insufficient or if a batch is past expiration.
4. **Stock Movement & Audit:** Upon dispensing, quantity is decremented from the batch, a `StockMovement` ledger row is written, and medication charges are added to the patient's billing docket.

Workflow 3: Procurement & Supply Chain Lifecycle

1. **Purchase Requisition:** Department Head identifies low stock of surgical gloves and submits a Purchase Requisition.
2. **Supplier Quotations:** Procurement Officer invites quotations from approved vendors and compiles a comparative price statement.
3. **Purchase Order (PO):** Hospital Admin authorizes the Purchase Order with separated approval duty.
4. **Goods Receipt Note (GRN):** Warehouse receiving dock inspects delivered items, enters batch numbers and expiries, and logs damaged goods rejections.
5. **Accounts Payable (AP):** Accounts converts the verified GRN into a vendor invoice and schedules electronic payment.

SECTION 6: Hospital Financial Operations & Pricing Architecture

Hospital financial management requires zero tolerance for errors. Our system combines automated patient billing with double-entry general ledger accounting:

Revenue & Billing Category	Pricing Model & Configured Rate	Financial Accounting Posting Rule
General OPD Consultation	■500 per visit (Standard tariff)	Debit: Cash/Card Clearing Account Credit: OPD Consultation Revenue Account
Specialist / Surgeon Consultation	■800 – ■1,200 per visit	Debit: Cash/Card Clearing Account Credit: Specialist Consultation Revenue Account
Emergency Consultation	■1,000 (Triage rate)	Debit: Cash/Card Clearing Account Credit: Emergency Medical Revenue Account
General Ward Bed Rate	■1,500 per calendar day	Debit: Patient Inpatient Ledger Credit: Room & Nursing Revenue Account
Semi-Private Room	■2,500 per calendar day	Debit: Patient Inpatient Ledger Credit: Room & Nursing Revenue Account
Deluxe Private Suite	■4,500 per calendar day	Debit: Patient Inpatient Ledger Credit: Room & Nursing Revenue Account
Intensive Care Unit (ICU)	■7,500 per day (Telemetry included)	Debit: Patient Inpatient Ledger Credit: Critical Care ICU Revenue Account
Diagnostic Pathology Tests	■200 – ■1,500 per test code	Debit: Patient Billing Docket Credit: Pathology Diagnostic Revenue Account
Radiology & Imaging Scans	■800 (X-Ray) – ■6,500 (MRI)	Debit: Patient Billing Docket Credit: Radiology Imaging Revenue Account
Pharmacy Medicines	Configured MRP with batch tracking	Debit: Patient Billing Docket Credit: Pharmacy Sales Revenue; Inventory Asset reduced
Inpatient Advance Deposit	■5,000 – ■50,000 security deposit	Debit: Bank/Cash Clearing Account Credit: Patient Deposit Liability Account

The Principle of Double-Entry General Ledger:

Every single financial transaction in AURA HMS strictly posts equal debits and credits: Total Debits == Total Credits. When a patient pays ■500 cash for a consultation: **Debit:** Cash in Hand (Asset increases by ■500). **Credit:** OPD Consultation Income (Revenue increases by ■500). This architectural discipline ensures that the hospital Balance Sheet and Profit & Loss statement reconcile down to the exact rupee without manual book balancing.

SECTION 7: Technical Architecture Explained in Simple English

You do not need to be a software programmer to understand how this system works internally. Here is the plain-English translation of our technology stack:

Technology	What It Means in Plain English	Hospital Analogy
React + TypeScript (Frontend)	The visual screen interface you click and see in your web browser. TypeScript ensures that numbers, text, and dates are entered without typing mistakes.	The hospital reception desk, counters, and digital monitor displays.
Django REST Framework (Backend)	The powerful Python server engine that processes rules, enforces security, calculates bills, and validates medical policies.	The hospital administration office and medical superintendent verifying every rule.
Supabase PostgreSQL (Database)	A rock-solid relational database that permanently stores patients, medical histories, invoices, and audit logs with enterprise security.	The hospital strongroom where permanent legal health record binders are locked.
REST API (Application Interface)	The secure communication protocol connecting the browser screen to the server.	The hospital pneumatic tube or orderly carrying verified documents between departments.
RBAC (Role-Based Access Control)	Digital security locks preventing staff from accessing screens outside their job role.	Keycards: doctors open consulting cabins, cashiers open money drawers, nurses open medicine rooms.
RLS (Row Level Security)	Database-level security rules that guarantee Branch A staff cannot view or tamper with Branch B patient data.	Separated locked storage cabinets for different hospital branches.

SECTION 8: Comprehensive Implementation Audit & Disclosures

A senior architect must always present the exact, verifiable truth about what is live in the database versus what relies on simulated dockets:

Subsystem / Capability	Implementation Status	Technical Truth & Verifiable Evidence
Patient Registration & EMR	**FULLY VERIFIED**	Live PostgreSQL tables (`PatientProfile`, `Appointment`, `ConsultationNote`). Unique MRN generation and cryptographically signed clinical notes.
Inpatient Admissions & Beds	**FULLY VERIFIED**	Live `Ward`, `Bed`, `InpatientAdmission` tables. Real-time visual status transitions (Available, Occupied, Cleaning, Maintenance).
Central Tariff Synchronization	**FULLY VERIFIED**	Hospital Admin centralized pricing authority. Instant multi-branch synchronization with non-destructive historical billing preservation.
Pharmacy FEFO Dispensing	**FULLY VERIFIED**	Auto-selects nearest expiry batch. Prevents dispensing expired stock. Deducts live quantity and writes `StockMovement` audit logs.
General Ledger & P&L;	**FULLY VERIFIED**	Strict balanced double-entry vouchers (`Total Debits == Total Credits`). Real Profit & Loss derived from posted transaction vouchers.
HR Attendance Engine	**FULLY VERIFIED**	Directly queries `EmployeeAttendance` database records. Real clock-in/out timestamps, calculated duration, overtime, and zero dummy text.
Multilingual UI Synchronization	**FULLY VERIFIED**	100% dictionary parity across 5 languages (English, Hindi, Telugu, Tamil, Kannada) with instant live UI re-rendering.
External PACS/DICOM Hardware	**SIMULATION SUPPORTED**	Clinical findings and study UIDs are stored in DB; viewing raw DICOM imaging files uses simulated viewers since physical MRI machines are external.
Lab Analyzer Telemetry	**SIMULATION SUPPORTED**	An ASTM/HL7 packet simulator generates machine telemetry to demonstrate automated LIS ingestion when physical hematology analyzers are offline.
Razorpay Payment Gateway	**TEST-MODE VERIFIED**	Real Razorpay Orders API integration with HMAC-SHA256 signature verification operating in Test Mode with standard sandbox credentials.

SECTION 9: Live HMS Demonstration Script for Executive Presentations

Use this word-for-word presentation script when demonstrating the hospital application to leadership, management, or clients:

Step 1: Introduction (60 Seconds)

Speaker Script: "Good morning everyone. Today I am presenting AURA HMS, our enterprise hospital management platform. Our system automates the complete operational lifecycle of a multi-specialty hospital across 21 user roles and 27 clinical and financial modules. Every department—from the front-desk token queue to doctor consultation rooms, inpatient wards, diagnostic laboratories, pharmacy inventory, and double-entry accounting—is seamlessly connected in real time."

Step 2: Patient Registration & OPD Token Queue

Speaker Script: "Let us begin with a new patient, Rahul Sharma. At the reception desk, we register Rahul with one click. Notice how the system automatically issues a permanent Medical Record Number, MRN-2026-00105. We now book an appointment with Dr. Ananya. The system enforces our mandatory payment security gate: consultation fee ■500 is collected in cash. Once confirmed, Token #A-014 is generated and instantly broadcasted to our Digital Waiting Lounge TV display with an automated audio chime."

Step 3: Doctor Consultation & Clinical EMR

Speaker Script: "Logging in as Dr. Ananya, Rahul appears at the top of our OPD queue. We open his consultation. Here, the doctor logs vital signs, records Chief Complaints, and selects an ICD-10 diagnosis. Dr. Ananya prescribes an antibiotic and orders a Complete Blood Count lab test. Because Rahul has severe abdominal pain, she clicks 'Admit Patient' to recommend inpatient admission."

Step 4: Bed Allocation & Advance Deposit

Speaker Script: "Moving to the Admissions module, we view our interactive Bed Visualizer. We allocate Bed G-04 in the General Medical Ward at ■1,500 per day. The patient's family deposits a ■5,000 security advance. Instantly, Bed G-04 transitions from Green (Available) to Red (Occupied) across the hospital network."

Step 5: Nursing Station & Diagnostic Lab

Speaker Script: "At the Nursing Station, Sister Mary accesses her Medication Administration Record (MAR) and marks the antibiotic injection as administered. Simultaneously, in the diagnostic laboratory, technician Rajesh receives the specimen barcode, enters the CBC values, and the pathologist digitally authorizes the verified report, which appears immediately on Dr. Ananya's screen."

Step 6: 4-Point Discharge & Financial Settlement

Speaker Script: "When Rahul recovers, the system enforces our strict 4-Point Discharge Clearance: Doctor clinical clearance, Nursing custody clearance, Pharmacy return clearance, and Billing settlement. Total hospital charges equal ■3,400. Because the family deposited a ■5,000 advance, the system generates the tax invoice and automatically calculates a ■1,600 cash refund. Upon discharge, Bed G-04 automatically switches to Orange for housekeeping sanitization."

Step 7: Double-Entry Accounting & Management Oversight

Speaker Script: "Finally, in the Finance Suite, our cashier shift reconciles cash collections down to the rupee. The General Ledger automatically records balanced debits and credits, updating our live Profit & Loss statement. Branch Managers have a complete 360-degree monitoring cockpit with strict read-only controls to prevent record tampering."

SECTION 10: Beginner's Glossary of Hospital Terms

Hospital Term	Simple Everyday Meaning
OPD	**Outpatient Department:** For patients who visit for a consultation and go home the same day.
IPD	**Inpatient Department:** For patients admitted to a hospital bed for overnight observation or surgery.
MRN	**Medical Record Number:** The unique, permanent hospital identity card number assigned to a patient for life.
EMR	**Electronic Medical Record:** The digital chart storing patient illnesses, diagnoses, and prescriptions.
MAR	**Medication Administration Record:** The nursing log recording exactly which nurse gave which medicine at what time.
FEFO	**First Expiring, First Out:** Dispensing medicines that expire earliest first to eliminate pharmacy wastage.
ICD-10	**International Classification of Diseases:** Standardized global codes for medical diagnoses (e.g. A09 for gastroenteritis).
TPA	**Third Party Administrator:** The agency processing insurance authorizations between hospital and insurer.
Pre-Auth	**Pre-Authorization:** Advance written approval from an insurance company agreeing to cover a surgery bill.
Cashless	When the patient is treated without paying cash upfront because their approved insurance covers the bill.
PO & GRN	**Purchase Order & Goods Received Note:** The official order sent to a supplier and the verified warehouse receipt signed at the gate.
General Ledger	The master accounting book where every single financial debit and credit is permanently recorded.
RBAC	**Role-Based Access Control:** Digital permissions ensuring staff only access screens relevant to their job.
Discharge Summary	The official medical report given to a departing inpatient detailing treatments, tests, and home medicines.